

Gen AI Intern Technical Assignment

Submission Deadline: 24 Hours

Assignment Overview

Build a Document Question Answering Assistant that accepts a PDF, extracts its contents, retrieves relevant information using embeddings and similarity search, and answers user questions using an LLM. The solution must be implemented without high-level orchestration frameworks.

Functional Requirements

- Accept a PDF document as input.
- Extract readable text from the PDF.
- Split the extracted text into chunks (recommended: 500 characters with 50-character overlap).
- Generate embeddings using any suitable embedding model.
- Store embeddings in memory.
- Implement similarity search using cosine similarity with NumPy (no vector databases).
- Retrieve the top three relevant chunks for each user query.
- Send only the retrieved chunks to the LLM and return the generated answer.

Restrictions

- Do not submit AI-generated code without understanding it.
- Cursor/Windsurf agent mode or equivalent autonomous coding tools should not be used.

Bonus (Optional)

Implement conversation history so the assistant can maintain context across multiple user queries.

Submission Deliverables

- Complete source code.
- requirements.txt.
- README with setup and execution instructions.
- A sample PDF for testing.
- A 3–5 minute screen recording demonstrating the application, explaining the code structure and major implementation decisions.

AI Usage Policy

Candidates may use AI assistants to understand concepts, review documentation, or debug isolated issues. However, the submitted solution must reflect their own understanding. During the technical review, candidates should be able to explain and modify every part of their implementation.